

Critical strip-crossing mass on the honeycomb lattice

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Abstract

At the critical weight of the regular honeycomb lattice, the total weight of self-avoiding paths crossing a strip of height N is comparable to $N^{-1/4}$. The first horizontal-displacement moment of return paths is comparable to $N^{3/4}$.

1 Introduction

A self-avoiding path at its critical weight can make a long excursion without paying an exponential cost. A basic quantitative question is how much of its weight reaches the opposite side of a strip. On the regular honeycomb lattice we prove that this crossing mass has power $-1/4$, with upper and lower bounds by fixed positive constants. Our proof also determines the first displacement moment of paths that return to the starting side. This second observable provides the route to the crossing estimate.

1.1 The model and the result

We use the honeycomb lattice dual to the equilateral triangular tiling of side length one, with one triangular edge direction horizontal. The separation of consecutive horizontal band boundaries is $d_0 = \sqrt{3}/2$. A *port* is the midpoint of an edge on the boundary of a union of triangles. A path between distinct ports follows dual edges, including the two terminal half-edges, visits each dual vertex at most once, and meets the boundary only at its endpoints. Its length $|\gamma|$ is the number of visited dual vertices; ports carry no weight. We assign the critical weight $\rho^{|\gamma|}$, where

$$\rho = \frac{1}{\sqrt{2 + \sqrt{2}}}, \quad c = \cos(3\pi/8). \quad (1)$$

This vertex convention agrees with the usual edge convention for the connective constant up to the fixed endpoint factor. Duminil-Copin and Smirnov proved that the reciprocal of ρ is the honeycomb connective constant [9].

Let $\mathcal{S}_N$ be the infinite horizontal strip of N triangular bands, where $N \geq 1$. Index its bottom ports consecutively by $\mathbb{Z}$, fix the source at port 0, and write $Z_D(a, b)$ for the sum of critical weights of paths from a to b in D . A path returning to the bottom is called an *arch*; a path ending at the top is called a *bridge*. In both cases all other points of the path are in the interior of the strip. Define

$$\begin{aligned} K_N(k) &= Z_{\mathcal{S}_N}(0, k), \\ \mathcal{A}_N &= \sum_{k \neq 0} K_N(k), \quad \mathcal{B}_N = \sum_{b \text{ on the top}} Z_{\mathcal{S}_N}(0, b), \\ m_N &= \sum_{k \geq 1} k K_N(k). \end{aligned} \quad (2)$$

The sums run over actual admissible boundary ports. Adjacent bottom ports have horizontal separation one, so m_N is the first rightward horizontal-displacement moment. Top ports lie on the translated lattice $N/2 + \mathbb{Z}$. These are unnormalized masses: for example the bridge probability measure would divide each bridge weight by $\mathcal{B}_N$. Figure 1 shows the convention.

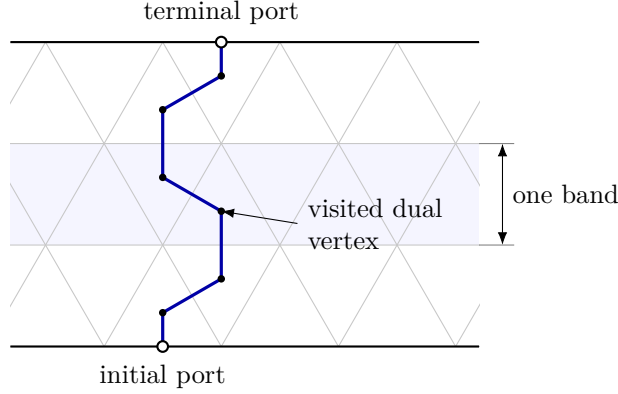


Figure 1: Port and length conventions. The path passes through six adjacent triangles of the triangular tiling, so it visits six vertices of the honeycomb dual and has port weight ρ^6 . Its two terminal pieces are half-edges. An ordinary vertex-to-vertex path counts full edges instead. The shaded region is one triangular band.

Theorem 1.1 (Critical strip mass). *For every integer $N \geq 1$, all sums in (2) are finite. With constants independent of N ,*

$$c\mathcal{A}_N + \mathcal{B}_N = 1, \quad m_{N+1} - m_N \asymp \mathcal{B}_N, \quad m_N \asymp N^{3/4}, \quad \mathcal{B}_N \asymp N^{-1/4}. \quad (3)$$

Moreover, $\mathcal{B}_N$ is nonincreasing in N .

Here $f_N \asymp g_N$ means that their ratio lies between two fixed positive constants. Thus the theorem determines a bounded-factor power law, rather than only a logarithmic exponent. If a height-zero term is useful, we set $\mathcal{B}_0 = 1$ by convention; no strip path is being added to the definition above. The crossing estimate can then be written $\mathcal{B}_N \asymp (1 + N)^{-1/4}$ for $N \geq 0$.

1.2 Earlier work

The two-dimensional polymer exponents arose from the dilute $O(n)$ model and its Coulomb-gas analysis. Nienhuis predicted the honeycomb critical point and the self-avoiding-walk exponents at $n = 0$ [22]. Lawler, Schramm and Werner developed the conjectural conformally invariant scaling picture for planar self-avoiding walk [21, Sections 3.3.1, 3.4.3 and 4.1]. Their boundary exponent $5/8$ predicts a boundary-to-boundary mass of power $-5/4$. Summing the endpoint along the opposite side of a strip then predicts the crossing power $-1/4$; this deduction is also stated explicitly in [9, Section 4].

Duminil-Copin and Smirnov proved the critical-point prediction using a parafermionic observable whose local cancellation gives a positive boundary identity [9, Lemmas 1 and 2]. Their strip argument gives bounds of order $1/N$ and 1 for the crossing mass. These bounds leave the predicted power undetermined.

The decay of this mass was subsequently proved by Beaton, Bousquet-Mélou, de Gier, Duminil-Copin and Guttmann in their study of surface adsorption [2, Theorem 10]. Glazman and Manolescu

obtained a shorter proof, a logarithmic bound along a sequence of heights, and invariance of the half-plane boundary two-point function under columnwise rhombic deformations with angles in $[\pi/3, 2\pi/3]$ [16, Proposition 1.1 and Theorems 1–2]. Krachun and Panagiotis proved a polynomial upper bound $\mathcal{B}_N \leq 100N^{-10^{-10}}$ in their strip convention, together with quantitative sub-ballisticity [20, Theorems 1–2]. Theorem 1.1 establishes the predicted strip power with uniform multiplicative constants in the port convention above. The exponent $3/4$ for m_N describes an unnormalized displacement moment summed over all path lengths, with strip height as its scale. The fixed-length displacement conjecture and the conjectured conformal scaling limit concern different observables.

1.3 Proof strategy

The boundary identity alone does not determine this power. Our route is to calculate m_N first and then recover $\mathcal{B}_N$ from its increments. The argument passes from finite strip transfers to a polynomial formula, then to a positive integral that can be estimated uniformly in the height.

In Section 2, cancellation of closed excursions gives the boundary identity and controls finite-height transfer matrices. We introduce rhombic weights with one spectral parameter in each row. A cut between columns records vacant positions and occupied positions paired by path pieces on one side of the cut. The stationary vacuum vector records the weighted sums of these partial pairings with no exterior source; a source vector has one additional strand connected to a specified boundary source. Their local relations belong to the dilute-loop integrable structure; the relation with discrete parafermions was developed by Ikhlef and Cardy, and the weighted-walk normalization and corrected general formulas were treated by Glazman [18, 15]. We prove the exact local identities used here in Appendix A.

Section 3 constructs a scalar Pfaffian p_N that turns the stationary vacuum into a vector of bounded-degree polynomials. A degree bound is essential: it makes interpolation an identity argument. Exchange relations and reductions at special parameters have this role in earlier dense and dilute loop calculations [8, 13, 14]. Here the degree and stationarity arguments are proved for zero loop weight. Section 4 cuts an arch at every column between its endpoints. This converts m_N into a pairing of source vectors and then into one coefficient of p_{N+1}/p_N . A stepped-boundary flux identity proves $m_{N+1} - m_N \asymp \mathcal{B}_N$.

Section 5 converts the homogeneous polynomial problem into a positive integral over $0 < y_1 < \dots < y_N < 1$, with density proportional to

$$\prod_{i=1}^N \omega(y_i) \prod_{i < j} \frac{(y_j - y_i)^2}{y_i + y_j}, \quad \omega(y) = \frac{1}{2} \sqrt{\sqrt{\frac{1+y}{2y}} - 1}.$$

Positivity of a Cauchy-moment determinant makes the coincident-parameter limit nonsingular. The pair interaction in the resulting integral is the one appearing in generalized Bures ensembles [12]. The derivation from the strip polynomial fixes the normalization and identifies the integral with the arch moment. Finally, Section 6 compares this measure with Bures–Laguerre laws. Schur’s Pfaffian identity and de Bruijn’s integration formula give their normalizing constants; positive association gives the comparison. The smallest-coordinate estimates yield $m_N \asymp N^{3/4}$. Monotonicity and the increment identity then give the crossing power in Theorem 1.1.

2 Positive flux and finite strip transfers

We write

$$t = \frac{3}{8}, \quad \lambda = \frac{\pi}{8}, \quad d = \cos(2\lambda) = \sin(2\lambda), \quad C = \cos \lambda. \quad (4)$$

Then $c = \cos(3\lambda)$ and $\rho = C - c = 1/(2C)$, consistently with (1).

The boundary identity controls positive path sums before any spectral parameter is introduced. We use it first to make the fixed-height transfer construction convergent. The local identities then allow the stationary vector to be recovered by polynomial interpolation.

Lemma 2.1 (Boundary identity). *Let D be a finite simply connected union of triangles with simple polygonal boundary, and let a be a boundary port. If $W(\gamma)$ is the total signed turning of a path, measured from its inward initial direction to its outward final direction, then*

$$\sum_{b \in \partial D} \sum_{\gamma: a \rightarrow b} \rho^{|\gamma|} e^{itW(\gamma)} = 1. \quad (5)$$

If D is convex, its unsigned boundary partition sum is at most $1/c$.

Proof. Send mass one along the inward half-edge at a . At each newly visited vertex split it between the two possible continuations, multiplying by $\rho e^{i\lambda}$ and $\rho e^{-i\lambda}$, respectively. The identity $2\rho \cos \lambda = 1$ conserves total mass. Stop a branch when it exits D or is about to enter a vertex already visited. This produces a finite tree because D has finitely many vertices.

For a stopped collision, the newly closed cycle is simple and has a disjoint incoming stem. The stem lies outside the cycle: it starts on ∂D and cannot cross the cycle before their first common vertex. Consequently the unused branch at that vertex is exterior to the cycle. The cycle turn there is $\pi/3$ in the counterclockwise orientation, rather than $-\pi/3$; the clockwise statement is reversed. The turn from the stem into the cycle replaces this closing turn by its negative. The two ways to traverse the cycle therefore have winding increments $4\pi/3$ and $-4\pi/3$ relative to the stem. They have the same unsigned weight and cancel, since $e^{it(4\pi/3)} + e^{-it(4\pi/3)} = 0$. Cycle reversal is an involution on all stopped collisions. The remaining terminal mass gives (5).

For a path ending at b , close it using the counterclockwise boundary arc from b to a . The turns at the two ports are $\pi/2$, so the turning theorem for this simple closed curve gives

$$W(\gamma) = \pi - \text{turn}_{\partial D}(b \rightarrow a).$$

For a convex boundary the latter turn lies in $[0, 2\pi]$. Thus $|W(\gamma)| \leq \pi$ and $\cos(tW(\gamma)) \geq \cos(3\pi/8) = c$. Taking real parts proves the last assertion. $\square$

We next introduce auxiliary spectral parameters. A column has N rhombi stacked along horizontal edges; in row i the other edge makes angle u_i/t with the horizontal. The same tiles make sense as combinatorial diagrams for complex u_i . Their four ports carry noncrossing partial pairings. Port occupancies must agree when tiles are glued, and every closed component has weight zero. Put

$$\begin{aligned} a(u) &= \cos(2u - 3\lambda), & L(u) &= \sin(2\lambda + u) \sin(3\lambda + u) = \frac{C + a(u)}{2}, \\ A(u) &= \frac{d \sin(3\lambda - u)}{L(u)}, & U(u) &= \frac{d \sin u}{L(u)}, \\ B(u) &= \frac{\sin u \sin(3\lambda - u)}{L(u)} = \frac{a(u) - c}{a(u) + C}, & & \\ E(u) &= \frac{\sin(3\lambda - u) \sin(2\lambda - u)}{L(u)}, & F(u) &= \frac{\sin(u - \lambda) \sin u}{L(u)}. \end{aligned} \quad (6)$$

These weights come from Nienhuis's integrable $O(n)$ construction [23], in the self-avoiding-walk normalization displayed in [16]; the general parafermionic and integrable local formulas are discussed

in [18, 15]. The empty tile has weight one. A single connection between opposite ports has weight B . A bottom-left or top-right turn has weight A ; the other two turns have weight U . The configurations with two turns of the first or second type have weights E and F , respectively. At $u = \lambda$ or 2λ , splitting along the short diagonal recovers two triangular tiles, with weight ρ per visited triangle. At $u = 0$ or 3λ the row is flat: each route continues deterministically through the adjacent left or right port, and the two disjoint routes can coexist with weight one.

The dilute diagram algebra and its Yang–Baxter operators have an algebraic construction in Grimm and Pearce [17]. Their distinction between gauge-invariant periodic partition functions and other gauge-dependent quantities is relevant here: we specify the caps, scalar factors and port ordering, and prove the identities for these choices.

We use the planar diagram algebra whose inputs and outputs are ordered slots, each vacant or occupied. Composition glues matching slots and annihilates a mismatch or a closed loop. Tensor product means juxtaposition; transpose reflects a diagram, interchanging inputs and outputs. The two-slot operator $R(v)$ has coefficients $1, A(v), B(v), U(v), E(v), F(v)$ on, respectively, the empty diagram, a single vertical connection, a single diagonal connection, a cap alone or cup alone, two vertical connections, and a cap together with a cup. There are two choices for a single vertical or diagonal connection, as shown in Figure 2.

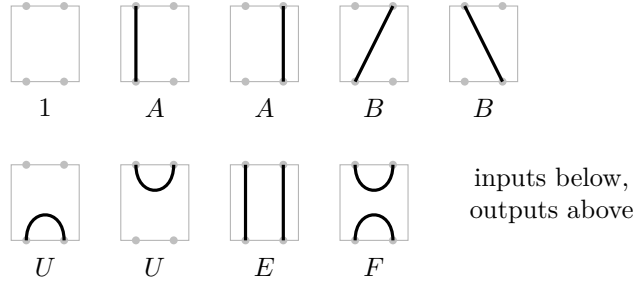


Figure 2: The nine local diagrams of $R(v)$, with coefficients evaluated at v . Grey dots mark available slots; a slot is occupied exactly when a black strand meets it. Gluing requires matching occupancies and assigns zero to every closed component. These are operator diagrams, rather than a geometric drawing of a rhombus.

Number the rows from bottom to top. A column swept from left to right uses $R(3\lambda - u_i)$ in row i , with local ordered inputs (south, west) and outputs (east, north). The north output of row i is the south input of row $i + 1$. The geometric and operator weights agree because

$$A(3\lambda - u) = U(u), \quad U(3\lambda - u) = A(u), \quad B(3\lambda - u) = B(u), \quad E(3\lambda - u) = F(u).$$

The last relation also gives $F(3\lambda - u) = E(u)$. Figure 3 records the port order.

Define S_2 , from one slot to two, by sending a vacancy to a vacancy plus ρ times a cup, and an occupied strand to either output slot with coefficient ρ . Define S_3 , from no slots to two, to be the sum of the vacant diagram and a cup.

Lemma 2.2 (Local diagram identities). *Writing R_i for action in slots $i, i + 1$, one has*

$$R_2(y)R_1(x + y)R_2(x) = R_1(x)R_2(x + y)R_1(y), \quad (7)$$

$$R(s\lambda) = S_s S_s^t, \quad s = 2, 3, \quad (8)$$

$$R_2(x)R_1(x + s\lambda)(I \otimes S_s) = (S_s \otimes I)\hat{R}_s(x), \quad (9)$$

$$R(x)R(-x) = I. \quad (10)$$

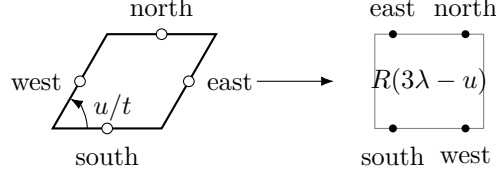


Figure 3: From a geometric rhombus to a column operator. South and west are the ordered inputs; east and north are the ordered outputs. The operator is drawn with its inputs below and outputs above, so the rhombic parameter u corresponds to the operator argument $3\lambda - u$. The rhombus is schematic; only the angle label and port order matter.

Here $\hat{R}_2(x) = R(x + \lambda)$ and $\hat{R}_3(x)$ is the one-slot identity. These are identities of rational functions. They can be specialized wherever the relevant sides are regular, including at removable singularities; an individual operator at a genuine pole is not assigned a finite value.

The complete coefficient proof is in Appendix A.

For a cut between two columns of N rows, number its slots from bottom to top. Let $\mathcal{L}_N^0$ be the vector space with basis the noncrossing partial pairings of occupied slots; all other slots are vacant. Let $\mathcal{L}_N^1$ have the same pairings with one additional unpaired occupied slot connected to an exterior source. This defect cannot be enclosed by a pair. Restricting a single path to one side of a cut can produce several paired fragments as well as the source-connected strand, as Figure 4 illustrates.

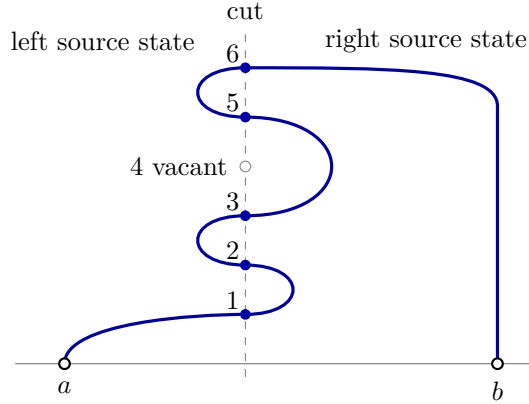


Figure 4: A single arch may cross a cut several times. In this schematic example the left half has a source strand at slot 1 and pairs (2, 3), (5, 6); the right half has pairs (1, 2), (3, 5) and a source strand at slot 6. Slot 4 is vacant on both sides. Each source strand is exposed, meaning that no pair encloses its slot. Gluing the two states recovers one path from a to b , with no closed component.

Before fixing its two auxiliary ends, the complete column is the map

$$M_X = R_N(v_N) \cdots R_1(v_1), \quad v_i = 3\lambda - u_i, \quad X = (u_1, \dots, u_N). \quad (11)$$

Its ordered inputs are $(\text{aux}, 1, \dots, N)$ and its ordered outputs are $(1, \dots, N, \text{aux})$: the auxiliary slot moves from the south end to the north end of the column. Attach a cut state to the west inputs, glue matching occupancies, and record the east state, assigning zero to any closed loop. With both auxiliary ends vacant, this defines $T_N^{(\epsilon)} : \mathcal{L}_N^\epsilon \rightarrow \mathcal{L}_N^\epsilon$, $\epsilon = 0, 1$; write $T_N = T_N^{(0)}$. With the south end occupied and the north end vacant, it defines the source insertion $\mathbf{b}_N : \mathcal{L}_N^0 \rightarrow \mathcal{L}_N^1$. Interchanging

these end conditions defines $\mathbf{b}_N^{\text{north}}$. A superscript $-$ denotes the corresponding maps for a sweep from right to left. Horizontal reflection replaces X by $X^* = (3\lambda - u_1, \dots, 3\lambda - u_N)$, so its complete column is $M_X^- = R_N(u_N) \cdots R_1(u_1)$. The empty coordinate of $T_N v$ equals that of v : any occupied input that closed up before reaching an output would form a forbidden loop.

Lemma 2.3 (Decay and rational continuation). *If every u_i belongs to $\{0, \lambda, 2\lambda, 3\lambda\}$, then T_N restricted to vectors with zero empty coordinate and $T_N^{(1)}$ have spectral radius strictly less than one. The same holds for reversed columns and in a complex neighborhood of each such parameter list, with absolute values of tile weights used to control absolute convergence. There is consequently a unique stationary vacuum vector P_N with empty coordinate one. Its entries are rational in the e^{iu_i} , and regular at every parameter list just specified.*

Proof. First take a homogeneous triangular parallelogram of height $H > 0$ and width M . The total weight of paths from a fixed left port to its right side tends to zero as $M \rightarrow \infty$. To see this, append a column immediately to the right of the exit and continue, entirely inside this new column, to its bottom port. The continuation has at most a fixed number of vertices depending only on H , so its weight is bounded below by a positive constant depending only on H . For every $M < L$ these extensions are paths in a parallelogram of width L . Their bottom endpoints identify M , and their first exit into the appended column recovers the original path. They are therefore distinct. Writing $q_H(M)$ for the crossing mass and C_H for the maximum number of added vertices, Lemma 2.1 gives

$$\rho^{C_H} \sum_{M < L} q_H(M) \leq 1/c.$$

Thus $q_H(M) \rightarrow 0$. The same lemma uniformly bounds paths joining ports on the same side.

For a mixed list of the four allowed parameters, collapse the flat rows. The row vectors belong to $\{1, e^{i\pi/3}, e^{2i\pi/3}, -1\}$, so outside two end pieces of horizontal size $O(N)$ the remaining triangles form a homogeneous parallelogram of height H , the number of nonflat rows, and width $M - O(N)$. This description does not depend on which neighboring triangles were grouped into rhombi. A collapsed deterministic route drifts by at most N columns. The end pieces have bounded size for fixed N and contribute bounded weights.

Every nonzero diagram contributing to either restricted transfer power has a component connecting the left and right cuts. In the vacuum sector, otherwise the nonempty input pairing would close into a loop; in the source sector, the source must reach the output. Cut all components at the two ends of the homogeneous core. There are only boundedly many endpoint patterns, depending on N , and at least one segment spans the core. Ignoring avoidance between these segments bounds their total weight by a product of single-path partition sums. Each nonspanning factor is at most $1/c$. For $H > 0$, choose the core width $M - \kappa_X$, where the fixed offset satisfies $0 \leq \kappa_X = O(N)$. If Q_N is the nonempty vacuum block and $\|\cdot\|_{\max}$ is the maximum absolute matrix entry, then

$$\|Q_N^M\|_{\max} + \|(T_N^{(1)})^M\|_{\max} \leq C_N \sum_{a \in \mathcal{E}_H} q_{H,a}(M - \kappa_X) \rightarrow 0.$$

Here $\mathcal{E}_H$ is the finite set of left ports of the core, $q_{H,a}$ is its crossing mass from a , and C_N absorbs the bounded end pieces and endpoint patterns. All constants are for fixed N ; no uniform spectral gap is needed. If $H = 0$, no spanning component is possible once M is sufficiently large. Thus both restricted matrix powers tend to zero. Since these are finite matrices, their spectral radii are less than one.

At the specified parameters all tile weights are nonnegative. The matrices obtained by taking absolute tile weights vary continuously, so their restricted spectral radii remain less than one in a

small complex neighborhood. This also proves the asserted absolute convergence. The reflected argument is identical.

Writing the vacuum matrix in the empty/nonempty decomposition gives $T_N = \begin{pmatrix} 1 & 0 \\ v & Q \end{pmatrix}$. Its normalized stationary vector is $(1, (I - Q)^{-1}v)$, which is regular in the stated neighborhoods and rational in the spectral variables. This proves both uniqueness and the continuation claim. Equivalently, P_N is the sum of finite path diagrams in the vacuum half-strip. $\square$

We now pass from local identities to column identities. This also records the source insertions needed for the arch moment. Let X' be X with adjacent rows $i, i+1$ exchanged, and put $\delta = u_{i+1} - u_i$. Applying the braid relation to those rows gives

$$M_X(I_{\text{aux}} \otimes R_i(\delta)) = (R_i(\delta) \otimes I_{\text{aux}})M_{X'}. \quad (12)$$

The tensor positions reflect the different input and output orders in (11). If $u_{i+1} = u_i + s\lambda$, $s = 2, 3$, let X_{red} replace these rows by $u_i + \lambda$ for $s = 2$, or delete them for $s = 3$. The splitter S_s acts on these cut slots, with identities on all other slots. In each sector it maps $\mathcal{L}_{N-r_s}^\epsilon$ to $\mathcal{L}_N^\epsilon$, where $r_2 = 1$ and $r_3 = 2$. The local splitting identity yields

$$M_X(I_{\text{aux}} \otimes S_s) = (S_s \otimes I_{\text{aux}})M_{X_{\text{red}}}. \quad (13)$$

Indeed its argument is $x = v_{i+1}$, and for $s = 2$ the effective box has argument $x + \lambda = 3\lambda - (u_i + \lambda)$, as required. Transpose splitting and reverse the slot order to obtain the reflected projection identity

$$(S_s^t \otimes I_{\text{aux}})M_X^- = M_{X_{\text{red}}}^-(I_{\text{aux}} \otimes S_s^t). \quad (14)$$

All three equations leave the auxiliary endpoints untouched. We can therefore impose either vacant ends or either source-insertion condition. With X as a subscript denoting its row list, this gives, for $\epsilon = 0, 1$,

$$\begin{aligned} T_X^{(\epsilon)} R_i(\delta) &= R_i(\delta) T_{X'}^{(\epsilon)}, & \mathbf{b}_X R_i(\delta) &= R_i(\delta) \mathbf{b}_{X'}, \\ T_X^{(\epsilon)} S_s &= S_s T_{X_{\text{red}}}^{(\epsilon)}, & \mathbf{b}_X S_s &= S_s \mathbf{b}_{X_{\text{red}}}, \\ S_s^t T_X^{(\epsilon), -} &= T_{X_{\text{red}}}^{(\epsilon), -} S_s^t, & S_s^t \mathbf{b}_X^- &= \mathbf{b}_{X_{\text{red}}}^- S_s^t. \end{aligned} \quad (15)$$

The same equations hold with $\mathbf{b}^{\text{north}}$ in place of $\mathbf{b}$. In a source-insertion equation the cut map on its right acts in the vacuum sector and the cut map on its left acts in the one-source sector.

The normalized stationary vector is now determined under exchange and reduction:

$$P_N(\dots, u_i, u_{i+1}, \dots) = R_i(u_{i+1} - u_i)P_N(\dots, u_{i+1}, u_i, \dots), \quad (16)$$

$$P_N = S_s P_{\text{red}} \quad \text{when } u_{i+1} = u_i + s\lambda. \quad (17)$$

The column equations prove stationarity of the right sides. Their empty coordinate is one: a cap producing an empty output from a nonempty pairing would close a zero-weight loop. Uniqueness in Lemma 2.3 proves exchange near the all-zero parameter list, and reduction near a mixed list with $u_i = 0$. Intersecting these regular neighborhoods with the specialization hypersurface gives an open set in that hypersurface. Rational continuation then proves the identities for general parameters. The reflected projection and source equations will be used in the same way, with uniqueness of the corresponding stationary or inhomogeneous transfer equation.

At $u_1 = 3\lambda$, the first slot is vacant and can be deleted. At $u_1 = 0$, the part with first slot vacant is P_{N-1} ; the occupied part is obtained from P_{N-1} by one column on rows $2, \dots, N$ with a south source and vacant north, tying that source to slot 1. These assertions follow from $R(0) = I$ and

$R(3\lambda) = S_3 S_3^t$; in the latter identity an occupied first input with vacant auxiliary input contributes zero. Finally, shifting u_k by π multiplies each tile weight by the occupancy signs at its west and east ports. Thus it conjugates T_N by the diagonal map D_k that changes the sign of states with slot k occupied. Since D_k preserves the empty coordinate, uniqueness gives $P_N(X + \pi e_k) = D_k P_N(X)$, where e_k is the k th coordinate vector.

3 The polynomial vacuum

The stationary vacuum is initially only a rational vector. To extract a physical observable at coincident parameters, we need a polynomial numerator with a controlled degree. We first construct its scalar normalization and then prove the vector degree bound by interpolation and stationarity.

Exchange equations, degree bounds and reductions at special rapidities have earlier uses in the dense $O(1)$ model [8] and in the open dilute $O(1)$ model [13, 14]. The latter comparison concerns the fusion construction of the polynomial normalization. A related verification of a Pfaffian normalization by deletion identities and degree bounds appears for the crossing $O(1)$ model in [7, Section 2.6]. We construct the present zero-loop-value scalar and vector with their own degree and stationarity proofs.

Write $X = (u_1, \dots, u_N)$ and $a_i = a(u_i)$. Define

$$D(a, b) = a^2 + b^2 + 2dab - d^2, \quad (18)$$

$$p_N(a_1, \dots, a_N) = \prod_{i < j} \frac{D(a_i, a_j)}{a_i - a_j} \text{Pf} \left[\frac{(a_i - a_j)(a_i + a_j + \rho)}{D(a_i, a_j)} \right].$$

For odd N , augment the skew matrix by a last column of ones and its negative transpose; set $p_0 = 1$.

Lemma 3.1 (Scalar reduction identities). *The function p_N is a symmetric polynomial of degree at most $N - 1$ in each argument. Its leading coefficient in any argument is p_{N-1} of the remaining arguments, and*

$$p_N(Y, c) = p_{N-1}(Y) \prod_{z \in Y} (z + C). \quad (19)$$

For $f_j = \cos(w + j\lambda)$,

$$p(Y, f_0, f_6) = p(Y)(f_0 + f_6 + \rho) \prod_{z \in Y} (z - f_{-6})(z - f_{12}), \quad (20)$$

$$p(Y, f_{-2}, f_2) = p(Y, f_0)(f_{-2} + f_2 + \rho) \prod_{z \in Y} (z + f_0). \quad (21)$$

Here the subscript of p is its number of arguments. Both formulas also hold with all shifts negated.

Proof. After multiplication by all D factors, the Pfaffian is an alternating polynomial, hence divisible by the Vandermonde product in (18). Counting degrees before and after division gives degree at most $N - 1$ in each variable. Letting a last argument tend to infinity proves the leading-coefficient statement: its kernel entries tend to -1 . In odd size first add the augmenting column and row to that argument's column and row. At the flat value, $D(z, c) = (z - c)(z + C)$ and the kernel entry equals one. This proves (19).

At $D(f_0, f_6) = 0$ only Pfaffian terms pairing these two arguments survive after denominators are cleared. The identity $D(f_0, z) = (z - f_{-6})(z - f_6)$ then gives (20).

Prove the merge identity by induction on $r = |Y|$. The case $r = 0$ follows from $p_2(a, b) = a + b + \rho$. If $r \geq 1$, distinguish a spectator z . Both sides have degree at most $r + 1$ in z , and their leading

coefficients agree by induction. They vanish at $z = -f_0$: delete this argument with f_{-2} , and the remaining factor involving f_2 vanishes. At $z = c$ they agree by the flat formula and

$$(C + f_{-2})(C + f_2) = (c + f_0)(C + f_0).$$

They also agree at each deletion value of z with another spectator. To check the factors, write that pair as g_0, g_6 , where $g_j = \cos(v + j\lambda)$, and use

$$\prod_{\ell=\pm 2} (f_\ell - g_{-6})(f_\ell - g_{12}) = (f_0 + g_0)(f_0 + g_6)(f_0 - g_{-6})(f_0 - g_{12}).$$

This is the product-to-sum identity obtained from shifts by $\pm 2\lambda$. There are $2(r-1)$ such deletion values, in addition to $-f_0$ and c . For generic arguments these are distinct and give at least $r+1$ roots for a difference of degree at most r , after its leading coefficient has canceled. Polynomial continuation concludes the proof. $\square$

Lemma 3.2 (Vacuum degree bound). *The vector $W_N = p_N P_N$ is Laurent polynomial in $e^{iu_1}, \dots, e^{iu_N}$, with each exponent between $-(2N-2)$ and $2N-2$. In the last variable $y = u_N$, $b = a(y)$, its last-slot-vacant part is a polynomial in b of degree at most $N-1$ with leading coefficient W_{N-1} . Its last-slot-occupied part is $\sin y$ times a polynomial in b of degree at most $N-2$. Set $W_0 = 1$.*

Proof. The assertion for $N = 1$ is immediate. We construct the asserted polynomial by induction and then prove its stationarity. Throughout the construction, W denotes the candidate at size N , whereas every smaller $W_j = p_j P_j$ is already known by induction. We first remove the candidate's possible poles and bound its degree, then establish its special-row relations, and finally use those relations to prove stationarity.

Deleting a vacant last slot identifies that sector with $\mathcal{L}_{N-1}^0$. Deleting an occupied last slot leaves its partner as the exposed defect of a state in $\mathcal{L}_{N-1}^1$. We use these identifications for the subscripts v and o below.

Construction and regularity. For each earlier row $i < N$, introduce the deletion node and its associated cosine values

$$y_i = u_i + 3\lambda, \quad \beta_i = a(y_i), \quad \gamma_i = a(u_i - 3\lambda), \quad \sigma_i = a(u_i + 6\lambda).$$

Let $\mathcal{R}_i$ move the newly inserted row y_i from immediately above row i to the last position, using successive boxes with arguments $y_i - u_j$, $i < j < N$. By (16), (17), and (20), the value required at $y = y_i$ is

$$Y_i = (a_i + \beta_i + \rho) \prod_{\substack{j < N \\ j \neq i}} (a_j - \gamma_i)(a_j - \sigma_i) \mathcal{R}_i S_3 W_{N-2}(X \setminus \{u_i, u_N\}). \quad (22)$$

These data agree with the genuine rational vector $p_N P_N$ on the node hypersurface: one may check the identity near $u_i = 0$ with the other rows mixed, where all stationary vectors are regular. The mixed point supplies an open neighborhood within the hypersurface, so this establishes a rational identity in the remaining parameters, not just equality at one point.

For distinct nodes put

$$\ell_i(b) = \prod_{\substack{j < N \\ j \neq i}} \frac{b - \beta_j}{\beta_i - \beta_j}.$$

Define the candidate by the two interpolation formulas

$$\begin{aligned} W_v(b) &= W_{N-1} \prod_{i < N} (b - \beta_i) + \sum_{i < N} (Y_i)_v \ell_i(b), \\ W_o(y) &= \sin y \sum_{i < N} \frac{(Y_i)_o}{\sin y_i} \ell_i(b). \end{aligned} \tag{23}$$

The vacant part has degree at most $N - 1$ and leading coefficient W_{N-1} ; the occupied part divided by $\sin y$ has degree at most $N - 2$. Both have the prescribed node values. Scalar interpolation shows that the empty coordinate of W is p_N .

We check carefully that this construction has no poles in earlier spectral variables. Each moving box has two sine factors in its denominator; they divide the factors $(a_j - \gamma_i)(a_j - \sigma_i)$ in (22). The possible divisor $\sin y_i$ in the occupied data also cancels. Explicitly, for $x = u_i$ and $v = u_j$, the two cancellations are

$$\begin{aligned} \frac{(a(v) - a(x - 3\lambda))(a(v) - a(x + 6\lambda))}{L(x + 3\lambda - v)} &= -4 \sin(x + v - 6\lambda) \sin(x + v + 3\lambda), \\ a(x) + a(x + 3\lambda) + \rho &= 4c \sin(x + 3\lambda) \sin(3\lambda - x). \end{aligned}$$

The remaining possible divisors come from colliding interpolation nodes. Since

$$\beta_i - \beta_j = -2 \sin(u_i + u_j + 3\lambda) \sin(u_i - u_j),$$

the two components of $\beta_i = \beta_j$ are

$$u_i = u_j \pmod{\pi}, \quad u_i + u_j = -3\lambda \pmod{\pi}.$$

On the first component the data, including the divided occupied data, agree by their identities with the true vector and its occupancy signs. This can be verified near the regular mixture $u_i = u_j = 0$, $y_i = y_j = 3\lambda$. On the second component both data vanish through their factors involving σ_i, σ_j ; at a generic point no moving-box denominator vanishes. These checks at generic points of every irreducible divisor suffice, since all possible denominators are products of the indicated simple Laurent factors. Thus W is Laurent polynomial in the earlier variables; multiple intersections of these divisors introduce no further denominators.

For its degree, send an earlier u_k to either Laurent extreme. The moving boxes remain bounded there, as follows directly from the weights in (6). Data with $i \neq k$ grow at most like a_k^{N-1} : they contain two scalar factors and the inductive vector W_{N-2} , while $\ell_i(b)$ stays bounded. Data with $i = k$ grow at most like $a_k^{1+2(N-2)}$, while their Lagrange cardinal polynomial contributes $O(a_k^{-(N-2)})$. The leading-coefficient term $W_{N-1} \prod_{i < N} (b - \beta_i)$ has the same bound. Since poles have already been removed, growth at the two Laurent extremes bounds every earlier exponent between $-(2N - 2)$ and $2N - 2$. Interpolation preserves the occupancy sign under a shift $u_k \mapsto u_k + \pi$; when the shift affects a node, its last-slot sign cancels the sign of the dividing sine.

Relations in the earlier rows. Earlier-row braid covariance follows from the same last-variable interpolation: the exchange box is independent of the last variable, the leading coefficients agree by induction, and every node datum agrees by the rational vacuum exchange identity. Uniqueness of the interpolant then gives covariance.

We have established the degree bounds and prescribed node values of W . To prove stationarity, we need more zeros than the last-variable interpolation supplies. The next two row specializations provide those zeros; the final argument combines them with the degree bounds and occupancy parity. Both specializations are proved by interpolation in the last variable.

1. If adjacent earlier rows satisfy $u_{r+1} = u_r + s\lambda$, $s = 2, 3$, the interpolant equals the smaller split vector multiplied by the corresponding scalar quotient from (20) or (21). The quotient has degree one in b for merge and two for deletion, with monic last-variable factor. Thus the required degree forms and leading coefficient agree by induction. At a node indexed by $i \neq r + 1$, compare both expressions with the true vector near a regular simultaneous specialization: take $u_r = 0$, $u_i = 0$ when distinct, and $y_i = 3\lambda$, with the other rows mixed. At $i = r + 1$ both expressions vanish. On the split side the vanishing factor is $b - \sigma_r$ in deletion or $b + a(u_r + \lambda)$ in merge. On the node-data side it is $a_r - \gamma_i$ or $a_r - \sigma_i$, respectively; no moving box crosses row r . This proves the specialization.
2. At $u_1 = 0$ or 3λ , use the first-slot formulas following (17), replacing their smaller vacuum vector by W_{N-1} and multiplying by $\prod_{j=2}^N (a_j + C)$. The only degree check requiring attention is the south-source column in the occupied first-slot part. With north vacant, the last box contributes 1 or $A(y)$ to a vacant last output, from vacant or occupied input respectively, and $U(y)$ or $B(y)$ to an occupied last output. Multiplication by $b + C$ clears its denominator and gives exactly the required last-variable forms. Only the contribution 1 has the leading vacant degree, leaving the formula with the last row omitted. At the nodes the expressions agree at regular mixed specializations, taking $u_i = 0$ for $i > 1$. The exception $u_1 = 3\lambda$, $i = 1$, $y_i = 6\lambda$ is handled by the vanishing factors $a(y_i) + C$ and $a_1 + \beta_1 + \rho$.

On each specialization the nodes are generically distinct and their dividing sines nonzero; all other cases follow by continuation.

Stationarity. Consider the polynomial stationarity residual

$$V = \left(\prod_{j=1}^N L(u_j) \right) (T_N - I)W.$$

At each last-variable node it vanishes, because the data came from the genuine stationary vector. To determine its degree, put $n = N - 1$, $q = W_o / \sin y$ and $\Lambda_n = \prod_{j < N} L(u_j)$. Besides the north-source map $\mathbf{b}_n^{\text{north}}$, use the map $\mathbf{C}_n^{\text{north}} : \mathcal{L}_n^1 \rightarrow \mathcal{L}_n^0$ obtained from a column with north occupied and south vacant by tying its north end to the exterior defect of the input state. The resulting output is a vacuum pairing; a diagram that closes a loop contributes zero. These maps and the transfers on the first n rows are independent of y .

The last box contributes 1 or $A(y)$ to a vacant output and $U(y)$ or $B(y)$ to an occupied output. Using $L(y) = (b + C)/2$ and $\sin y \sin(3\lambda - y) = (b - c)/2$ therefore gives the exact blocks

$$V_v = \frac{\Lambda_n}{2} \left((b + C)(T_n - I)W_v + d(b - c)\mathbf{C}_n^{\text{north}}q \right), \quad (24)$$

$$\frac{V_o}{\sin y} = \frac{\Lambda_n}{2} \left((b - c)T_n^{(1)}q - (b + C)q + 2d\mathbf{b}_n^{\text{north}}W_v \right). \quad (25)$$

The second block has degree at most $N - 1$. In the first, the only possible degree- N term has coefficient

$$[b^N]V_v = \frac{\Lambda_n}{2}(T_n - I)W_n = 0$$

by the inductive stationarity. Thus V_v also has degree at most $N - 1$. Consequently every coordinate of V contains $\prod_{i < N} (b - \beta_i)$. In each earlier variable u_k its Laurent exponents are bounded by $2N$. The two sets of specializations just proved make V vanish when an earlier row is 0 or 3λ , or when

$u_j = u_i + s\lambda$, $i < j < N$, $s = 2, 3$. To pass from adjacent to arbitrary pairs, move the two rows next to one another without interchanging them, using earlier-row covariance and the column identity; generic invertibility follows from (10).

For fixed $k < N$, these zeros force divisibility by

$$(b - \beta_k) \sin u_k \sin(u_k - 3\lambda) \prod_{\substack{j < k \\ s \in \{2,3\}}} \sin(u_k - u_j - s\lambda) \prod_{\substack{k < j < N \\ s \in \{2,3\}}} \sin(u_j - u_k - s\lambda).$$

For generic other variables this divisor has Laurent extremes $\pm 2N$: the factor $b - \beta_k$ contributes two in each direction, the two flat sine factors contribute two, and the $2(N - 2)$ pair factors contribute the remaining $2N - 4$. Its parity is even under $u_k \mapsto u_k + \pi$. The quotient of a Laurent polynomial with the same exponent bound by this divisor is independent of e^{iu_k} . If slot k is occupied, however, the corresponding coordinate of V has odd parity, so that coordinate must be zero. Every nonempty vacuum pairing has an occupied earlier slot, since it has at least two occupied slots. The empty residual is identically zero. Hence $V = 0$, so W is stationary. Its empty coordinate is p_N ; uniqueness at generic regular parameters gives $W = p_N P_N$. $\square$

4 An exact formula for the arch moment

We now reconnect the polynomial vacuum with a positive path observable. Cutting a rightward arch at every column between its endpoints counts it exactly as many times as its horizontal displacement. We use this pairing of source states to identify the arch moment with a polynomial coefficient. A stepped-boundary flux calculation then compares its successive increments with the bridge mass.

For a repeated rhombic column $X = (u_1, \dots, u_N)$ near a physical parameter list, let $K_X(k)$ be the bottom-to-bottom partition sum with displacement k , with exactly its two endpoints occupied on the exterior boundary and one connected path. Set $m(X) = \sum_{k \geq 1} k K_X(k)$. A cut can meet the same arch several times: either half then contains its source fragment together with paired fragments, as in Figure 4. Using the south-source column $\mathbf{b}_N$ from Section 2, put

$$H_N = (I - T_N^{(1)})^{-1} \mathbf{b}_N P_N, \quad (26)$$

and define H_N^- in the reflected strip. Here P_N supplies the paired fragments before the source column, while the resolvent sums the columns between that source and the cut. Let Glue be the bilinear pairing of one-source states that is one when their occupancies agree and gluing joins the two sources into a single component with no loops, and zero otherwise. Then

$$m(X) = \text{Glue}(H_N^-, H_N). \quad (27)$$

Indeed expanding the two resolvents places one source on either side of the cut. For endpoint separation k there are exactly k possible cuts between them. Lemma 2.3 justifies absolute convergence of the source sums and the vacuum sums in a neighborhood of every mixed parameter list. Thus m is rational in the spectral variables and regular at all such lists.

Related finite-strip current calculations use bilinear link-state pairings, fusion reductions and interpolation [6, 11]. Those $O(1)$ current formulas use an assumed cancellation of a normalizing factor or additional symmetry. For the present zero-loop-value arch moment, the source-sector bound proved below supplies the polynomial control needed for interpolation.

Proposition 4.1 (Arch formula). *Let $\eta = 2d - 1$ and*

$$g_X(z) = \frac{p_{N+1}(a_1, \dots, a_N, z)}{p_N(a_1, \dots, a_N)}. \quad (28)$$

Then g_X is monic of degree N for generic X , and

$$m(X) = \sum_{i=1}^N a_i - \eta[z^{N-1}]g_X(z). \quad (29)$$

The right side is understood by continuation at removable singularities.

Proof. We identify m from its row reductions and a separate degree bound of $2N$ for $p_N^2 m$ in each a_i . The source-sector limit supplies that bound; the scalar reductions then determine the coefficient in (29).

Symmetries and reductions. Moving an exchange box through both halves of (27), using (15), proves symmetry in the rows. At an ascending merge or deletion specialization, the same identities for the vacuum and south-source columns give

$$P_N = S_s P_{\text{red}}, \quad H_N = S_s H_{\text{red}}.$$

The reflected column identity (14) gives

$$S_s^t P_N^- = P_{\text{red}}^-, \quad S_s^t H_N^- = H_{\text{red}}^-.$$

For the first reflected equality, the projected vector is stationary with empty coordinate one. For the second, the projected source equation is

$$(I - T_{\text{red}}^{(1),-}) S_s^t H_N^- = \mathbf{b}_{\text{red}}^- S_s^t P_N^- = \mathbf{b}_{\text{red}}^- P_{\text{red}}^-. \quad (30)$$

Uniqueness near a mixed specialization and rational continuation therefore identify the projected vector with H_{red}^- . Transposing the splitter inside the gluing pairing now reduces m to the smaller strip. Thus the two halves reduce by insertion and projection, respectively; no identity $S_s^t S_s = I$ is required. Occupancy signs cancel under a π shift of a parameter.

There is also a separate reflection symmetry $u_j \leftrightarrow 3\lambda - u_j$. By exchange, put that row on top. Since all top ports are vacant, each visit to the row is an interval that turns up and then back down. An interval with displacement $\ell \geq 1$ has weight $AUB^{\ell-1}$, which is invariant under this reflection. The intervals are ordered and disjoint. The argument is valid for absolutely convergent sums near physical parameters and hence as a rational identity. If $a_j = c$, then $AU = 0$ and these excursions vanish, so the flat row can simply be omitted.

A polynomial bound from the source sector. To obtain the degree bound, first use a north source. Add a last row y to the vacuum strip, and identify its occupied last-slot sector with one-source states on the earlier rows. Last-box stationarity is

$$(P_{N+1})_o = B(y)T_N^{(1)}(P_{N+1})_o + U(y)\mathbf{b}_N^{\text{north}}(P_{N+1})_v. \quad (31)$$

As $b = a(y) \rightarrow \infty$, Lemmas 3.1 and 3.2 give $(P_{N+1})_v \rightarrow P_N$, while $B(y) \rightarrow 1$. Thus $(P_{N+1})_o/U(y)$ tends to the integrated north-source vector. Since $U(y) = 2d \sin y/(b + C)$, its numerator after multiplication by p_N is exactly

$$\frac{1}{2d}[b^{N-1}] \frac{(W_{N+1})_o}{\sin y}.$$

The lemma bounds its Laurent exponents in each earlier variable by $2N$. Reflection from top to bottom reverses row order and replaces u_i by $3\lambda - u_i$; it preserves p_N and these exponent bounds.

Hence both $p_N H_N$ and $p_N H_N^-$ satisfy the same bound. By (27), $p_N^2 m$ has exponent bound $4N$. Its separate reflection and π -periodicity imply that it is a polynomial in $a_i = \cos(2u_i - 3\lambda)$ of degree at most $2N$ in each variable. Indeed, a π -periodic Laurent polynomial is a Laurent polynomial in $z = e^{2iu_i}$, and invariance under $z \mapsto e^{6i\lambda}/z$ makes it a polynomial in the corresponding scaled sum a_i . Row exchange makes this polynomial symmetric.

Interpolation. Let the right side of (29) be $\tilde{m}$. The scalar identities give the exact quotient rules

$$g_{\text{merge}}(z) = (z + f_0)g_{\text{red}}(z), \quad g_{\text{delete}}(z) = (z - f_{-6})(z - f_{12})g_{\text{red}}(z), \quad g_{\text{flat}}(z) = (z + C)g_{\text{red}}(z).$$

The elementary identities

$$f_{-2} + f_2 - f_0 = \eta f_0, \quad f_0 + f_6 = -\eta(f_{-6} + f_{12}), \quad c = \eta C$$

show that $\tilde{m}$ has the same three reductions as m . Both $p_N^2 m$ and $p_N^2 \tilde{m}$ have degree at most $2N$ in each a_i . Induct on N . In one distinguished variable, every other argument $a_j = \cos \theta_j$ supplies the two merge values $\cos(\theta_j \pm 4\lambda)$ and the two deletion values $\cos(\theta_j \pm 6\lambda)$, in addition to the flat value c . These are $4N - 3$ distinct roots for generic other arguments, more than $2N$ for every $N \geq 2$. Therefore the polynomial difference vanishes. The initial cases are $m(\emptyset) = 0$ and

$$m(u) = \frac{A(u)U(u)}{(1 - B(u))^2} = \frac{a(u) - c}{1 + d} = a(u) - \eta(a(u) + \rho).$$

In the size-zero quotient convention use $g_0 = 1$ and $[z^{-1}]g_0 = 0$. This completes the induction. $\square$

At $X = (\lambda, \dots, \lambda)$, each rhombic row splits into the two triangular tiles of one band, with weight ρ per visited vertex. Thus $a_i = C$ and $m(X) = m_N$ in the port convention of the introduction. In particular,

$$m_N = NC - \eta \lim_{a_1, \dots, a_N \rightarrow C} [z^{N-1}]g_X(z). \quad (32)$$

This coefficient has a removable limit because m is regular at the physical list. Section 5 will prove that the entire monic polynomial g_X has a unique limit and will evaluate this coefficient through a positive integral.

Lemma 4.2 (Moment increments and bridge mass). *For homogeneous honeycomb strips, $c\mathcal{A}_N + \mathcal{B}_N = 1$, the bridge mass is nonincreasing, and $m_{N+1} - m_N \asymp \mathcal{B}_N$ uniformly in $N \geq 1$.*

Proof. Apply Lemma 2.1 to parallelogram truncations of the strip. The left and right boundary terms tend to zero by Lemma 2.3: after the source column, propagate in the one-source sector to the distant cut, where the exit is the only occupied boundary port. The remaining vacuum states are bounded. The other sums increase to their infinite-strip limits. Bottom exits have winding $\pm\pi$ and top exits winding zero, so the real part is $c\mathcal{A}_N + \mathcal{B}_N = 1$. Increasing strip height increases the arch sum by inclusion, hence decreases $\mathcal{B}_N$.

Add one triangular row above the right half of $\mathcal{S}_N$, in columns $1, 2, \dots$, creating a stepped strip with a single side port p at the left end of the new row. Denote its bottom arch kernel by $\overline{K}(i, j)$; see Figure 5.

From a bottom source i , the imaginary part of the boundary identity is

$$\sin(3\lambda) \left(\sum_{j < i} \overline{K}(i, j) - \sum_{j > i} \overline{K}(i, j) \right) + \sin \lambda Z(i, p) = 0. \quad (33)$$

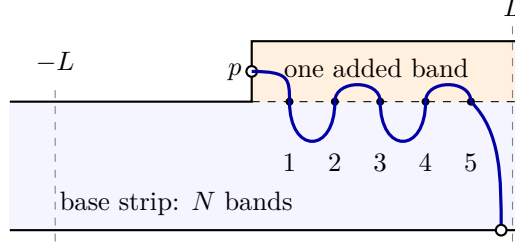


Figure 5: The stepped strip used to compare m_{N+1} and m_N . The drawing is schematic: curves represent path pieces, not individual lattice edges. Interface visits are numbered from left to right. A single connected path forces alternating consecutive pairings in the added row and the base strip; the final visit connects to the bottom. Moving the two dashed cuts to infinity recovers the two homogeneous arch moments.

Indeed horizontal top exits have winding zero, and closing along the leftward boundary gives winding $\pi/3$ at p . Lateral truncation errors vanish as before, dominated by a strip of height $N + 1$.

Sum (33) over $-L \leq i \leq L$. Reversal pairs and cancels all internal arches. The uncanceled terms cross the two ends of this interval. After translating each end to the origin, the stepped domains converge locally to strips of heights N and $N + 1$, respectively. All crossing sums are dominated by the first moment in the strip of height $N + 1$, which is finite by transfer decay. This also controls paths leaving an arbitrarily large lateral window for each fixed pair of ports. Dominated convergence therefore gives the exact identity

$$m_{N+1} - m_N = \frac{\sin \lambda}{\sin(3\lambda)} \sum_i Z(p, i). \quad (34)$$

Turning down immediately from p and following an arbitrary bridge in the base strip yields $\sum_i Z(p, i) \geq \rho \mathcal{B}_N$. The top-source bridge mass in the base strip equals its bottom-source mass by translation and reversal.

For the upper bound, order the path's visits to the interface between the base strip and the new row from left to right. In the new row, whose top is vacant, disjoint intervals pair p with the first visit and then pair subsequent visits consecutively. The base portions are a noncrossing pairing with one unpaired visit connected to the bottom; that defect cannot be enclosed. Since the union is a single path, the first base visit must pair with its next neighbor: pairing it farther away would leave a closed component inside. Removing these two visits and repeating proves that base visits also pair consecutively, with the last visit the defect. Thus the path encounters the interface in increasing horizontal order.

At $u = \lambda$, the initial new-row segment has total weight at most $A/(1 - B)$. A one-sided base return arch has total mass at most $1/(2c)$, using the boundary identity from the top and equality of the two directions by translation and reversal. A one-sided return through the new row has total mass $AU/(1 - B) = 2cB$. The final base segment has total mass $\mathcal{B}_N$. Ignoring avoidance between pieces only increases the sum, since the base weights are products over vertices. Therefore

$$\sum_i Z(p, i) \leq \frac{A}{1 - B} \sum_{r \geq 0} \left(\frac{1}{2c} 2cB \right)^r \mathcal{B}_N = \frac{A}{(1 - B)^2} \mathcal{B}_N.$$

Here $B = B(\lambda) = \rho^2 < 1$, so the constants are independent of N . Combine this with (34). $\square$

5 A positive integral for the homogeneous moment

The arch formula (29) is exact, but its Pfaffian quotient does not directly show the size of m_N . We convert the coalescing-parameter problem into a positive integral. This will establish the limiting quotient without assuming that its denominator remains nonzero, and make the moment accessible to comparison estimates. For generic X , put $b = \cos w$ and

$$Q_X(b) = (b - C)g_X(b) \prod_{i=1}^N (b - a_i)(\cos(w - 2\lambda) - a_i)(\cos(w + 2\lambda) - a_i), \quad (35)$$

$$\Phi_X(w) = \sin w Q_X(\cos w).$$

The paired shifted factors have product $b^2 - 2da_i b + a_i^2 - d^2$, so Q_X is a polynomial of degree $4N + 1$.

Lemma 5.1 (Missing harmonics). *Every sine harmonic of Φ_X whose frequency is divisible by four has coefficient zero. Equivalently, writing $h = b^2(1 - b^2) = (1 - \cos 4w)/8$, there are polynomials such that*

$$Q_X(b) = \mathcal{P}(h) + b\mathcal{J}(h) + b^2\mathcal{E}(h), \quad \deg \mathcal{P}, \deg \mathcal{J} \leq N, \quad \deg \mathcal{E} \leq N - 1. \quad (36)$$

Proof. Fix i and write $f_j = \cos(v + j\lambda)$ with $f_0 = a_i$. The product in (35) gives $\Phi_X(v \pm 2\lambda) = 0$. The deletion quotient also gives $\Phi_X(v + 6\lambda) + \Phi_X(v - 6\lambda) = 0$. To verify its sign and factors, cancel the contributions of the other a_j in the ratio: on both sides these contributions involve exactly $f_6, f_{-6}, f_4, f_{-4}, -f_0$. The remaining expressions are

$$R_{\pm} = \sin(v \pm 6\lambda)(f_{\pm 6} - C)(f_0 + f_{\pm 6} + \rho)(f_{\pm 6} - f_0)(f_{\pm 4} - f_0).$$

Product-to-sum and $\rho = 2cd$ give

$$R_{\pm} = \cos(2v)(f_{\pm 6} - C) \sin(v \pm 3\lambda)(\cos(v \pm 3\lambda) + d).$$

The last two expressions after removing $\cos(2v)$ are trigonometric polynomials of degree three. They have the same six zeros $v = \pm 3\lambda, \pm 5\lambda, \pm 7\lambda$ modulo 2π , and opposite values at zero; hence they are negatives of each other. This proves the claimed cancellation as a polynomial identity, including where canceled factors vanish.

Consider the sum of the four shifts of Φ_X by $\pm 2\lambda, \pm 6\lambda$. On frequency k its multiplier is $4 \cos(4k\lambda) \cos(2k\lambda)$, nonzero precisely when four divides k . Its maximum frequency is $4N + 2$. Dividing this shifted sum by $\sin 4w$ therefore gives a polynomial in $\cos 4w$ of degree at most $N - 1$. The preceding identities give N distinct roots, for generic a_i , so it is zero. This proves the harmonic assertion.

Every polynomial in b has a unique expression $P_0(h) + bP_1(h) + b^2P_2(h) + b^3P_3(h)$, by repeatedly using $b^4 = b^2 - h$. The first three terms, after multiplication by $\sin w$, have no frequency divisible by four. For the last term use

$$\sin w b^3 = \frac{1}{4} \sin 2w + \frac{1}{8} \sin 4w.$$

Its projection onto multiples of four is $\frac{1}{8} \sin 4w P_3((1 - \cos 4w)/8)$, so it vanishes only if $P_3 = 0$. The degree bounds follow from $\deg Q_X = 4N + 1$. $\square$

At the homogeneous parameters $a_i = C$, the paired factors in (35) become $(b - C)(b - c)$. Thus the desired limiting monic polynomial is characterized by

$$Q(b) = (b - C)^{2N+1}(b - c)^N g(b) = \mathcal{P}(h) + b\mathcal{J}(h) + b^2\mathcal{E}(h), \quad \deg g = N. \quad (37)$$

Lemma 5.2 (Nonsingularity and integral representation). *The monic problem (37) has a unique solution, which is the limit of g_X as all $a_i \rightarrow C$. Let*

$$\omega(x) = \frac{1}{2} \sqrt{\sqrt{\frac{1+x}{2x}} - 1}, \quad 0 < x < 1, \quad (38)$$

and let $(y_1, \dots, y_N)$ have density proportional to

$$\prod_{i=1}^N \omega(y_i) \prod_{i < j} \frac{(y_j - y_i)^2}{y_i + y_j}, \quad 0 < y_1 < \dots < y_N < 1. \quad (39)$$

Then

$$m_N = \frac{\eta}{\pi} \int_0^1 \left(1 - \mathbb{E} \prod_{i=1}^N \frac{y_i - x}{y_i + x} \right) \frac{\omega(x)}{x} dx. \quad (40)$$

Proof. We first turn the zero-multiplicity conditions into Cauchy-transform moment equations. A positive determinant will make those equations uniquely solvable and control the coalescing limit. We then identify the resulting coefficient with the expectation in (40). The proposed probability law is well defined: $\omega(x) = O(x^{-1/4})$ at zero and each pair factor is at most $y_j \leq 1$, so its normalizing integral is finite; it is positive because the density is positive inside the ordered region.

Cauchy moment equations. The two prescribed zeros satisfy $h(C) = h(c) = 1/8$. To place them at infinity, set $z = (8h - 1)^{-1}$, so that the inverse relation is

$$b^2(1 - b^2) = \frac{1 + 1/z}{8}.$$

We use the two branches near infinity tending to C and c , respectively. Their differences from these limits have order z^{-1} , since $h'(C)$ and $h'(c)$ are nonzero. For each polynomial in (37), multiply by z^N after the substitution $h = (1 + 1/z)/8$. Denote the resulting polynomials by $\mathcal{P}$, $\tilde{\mathcal{J}}$, and $\tilde{\mathcal{E}}$; the last satisfies $\tilde{\mathcal{E}}(0) = 0$ because $\deg \mathcal{E} \leq N - 1$. The branch tending to C and the sum of the two branches are, respectively,

$$f(z) = \sqrt{\frac{1 + \sqrt{(1 - 1/z)/2}}{2}}, \quad l(z) = \sqrt{1 + \sqrt{(1 + 1/z)/2}}.$$

These formulas continue f analytically off $[0, 1]$ and l off $[-1, 0]$. On the C branch, the zero of order $2N + 1$ in (37) becomes $O(z^{-2N-1})$; on the c branch, the zero of order N becomes $O(z^{-N})$. After multiplication by z^N this gives

$$\mathcal{F}(z) = \tilde{\mathcal{P}}(z) + f(z)\tilde{\mathcal{J}}(z) + f(z)^2\tilde{\mathcal{E}}(z) = O(z^{-N-1}), \quad (41)$$

$$\mathcal{D}(z) = \tilde{\mathcal{J}}(z) + l(z)\tilde{\mathcal{E}}(z) = O(1). \quad (42)$$

For the second statement, evaluate the same polynomial expression on the branch tending to c , where multiplication by z^N makes it $O(1)$, and subtract the expression on the C branch; the quotient by their nonzero difference is $\mathcal{D}$.

With subscripts $+$ and $-$ denoting boundary values from the upper and lower half-planes, the jumps across the cuts are

$$f_+(x) - f_-(x) = 2i\omega(x), \quad l_+(-x) - l_-(-x) = -2\sqrt{2}i\omega(x), \quad f_+(x) + f_-(x) = l(x).$$

At the outer endpoints these functions are bounded. At zero, $\mathcal{F} = O(|z|^{-1/4})$ and $\mathcal{D}(z) \rightarrow D_0 := \tilde{\mathcal{J}}(0)$; the latter uses $\tilde{\mathcal{E}}(0) = 0$. Cauchy's formula, with small endpoint circles shrinking to zero, yields

$$\mathcal{F}(z) = \frac{1}{\pi} \int_0^1 \frac{\mathcal{D}(x)\omega(x)}{x-z} dx, \quad (43)$$

$$\int_0^1 x^j \mathcal{D}(x)\omega(x) dx = 0, \quad 0 \leq j < N, \quad (44)$$

$$\mathcal{D}(z) = D_0 - \frac{\sqrt{2}}{\pi} z \int_0^1 \frac{\tilde{\mathcal{E}}(-y)/y}{z+y} \omega(y) dy. \quad (45)$$

The moment equations are the first N coefficients of the expansion of (43) at infinity. To obtain (45), subtract the bounded-at-infinity representation at zero. Its integrand is integrable there because $\tilde{\mathcal{E}}(-y)/y$ is a polynomial.

To control the endpoint at zero, use $\omega(x) = O(x^{-1/4})$ and $\tilde{\mathcal{E}}(z) = O(z)$ near zero. The increment in (45), split at $y = x$, is bounded by

$$C \left(\int_0^x y^{-1/4} dy + x \int_x^1 y^{-5/4} dy \right) = O(x^{3/4}).$$

This estimate will also justify the singular moment integral below.

Nonsingularity and coalescence. The polynomial $\tilde{\mathcal{E}}(-y)/y$ has degree at most $N-1$, and the moment equations determine it uniquely from D_0 . Indeed the corresponding matrix, apart from nonzero scalar factors, is

$$M_{jk} = \int_0^1 \int_0^1 \frac{x^{j+1}y^k}{x+y} \omega(x)\omega(y) dx dy, \quad 0 \leq j, k < N. \quad (46)$$

Its determinant is strictly positive. This is the Cauchy-bimoment positivity argument used in [3, Theorem 2.1]; we give the finite determinant calculation for the present weights. Expand the determinant and antisymmetrize successively in the x and y variables. On the ordered integration regions the integrand becomes the product of two positive Vandermonde determinants, $\prod_i x_i$, positive weights, and the Cauchy determinant

$$\det \left[\frac{1}{x_i + y_j} \right] = \frac{\prod_{i < j} (x_j - x_i)(y_j - y_i)}{\prod_{i,j} (x_i + y_j)} > 0.$$

For completeness this identity follows by multiplying through by the denominator: the numerator is alternating in each family and has exactly the degree of the two Vandermondes. The remaining constant is one, recursively evaluating at $x_N = -y_N$.

Consider now a solution of the homogeneous linear problem for g , so $\deg g \leq N-1$. The coefficient of b^{4N+1} in Q is zero. Since

$$[b^{4N+1}]Q = (-8)^N D_0,$$

we obtain $D_0 = 0$. Positivity of (46) forces $\tilde{\mathcal{E}} = 0$; then (45) gives $\tilde{\mathcal{J}} = 0$, and (41) gives $\tilde{\mathcal{P}} = 0$. Thus $g = 0$. For the monic problem, write $g(b) = b^N + \sum_{r=0}^{N-1} \gamma_r b^r$. The excluded $b^3 P_3(h)$ part in the proof of Lemma 5.1 has N coefficients. Requiring them to vanish therefore gives an $N \times N$ linear system for $(\gamma_0, \dots, \gamma_{N-1})$, and the established injectivity proves that this system is invertible at the homogeneous parameters. This proves existence and uniqueness of g . Apply the same coefficient conditions to (35), with g_X replaced by a monic unknown polynomial. Their matrix and right side

depend polynomially on the a_i , through the displayed zero factors, and converge to those of the homogeneous system. This coefficient matrix is consequently invertible for all a_i sufficiently close to C . The generic g_X satisfies that system by Lemma 5.1, so it converges to g . The moment itself is regular at the homogeneous point by Lemma 2.3.

The moment coefficient and its expectation. For this monic solution $D_0 = (-8)^{-N}$. Put $G(z) = \mathcal{D}(z)/D_0$. The leading coefficient computation also gives $[b^{4N}]Q = (-8)^N \tilde{\mathcal{P}}(0)$. Comparing this with the product in (37) gives the coefficient needed by the arch formula:

$$[b^{N-1}]g(b) = \frac{\tilde{\mathcal{P}}(0)}{D_0} + (2N+1)C + Nc.$$

Substitution into (32), using $C = \eta(2C + c)$, yields the first equality below:

$$m_N = -\eta \left(C + \frac{\tilde{\mathcal{P}}(0)}{D_0} \right) = \frac{\eta}{\pi} \int_0^1 (1 - G(x)) \frac{\omega(x)}{x} dx. \quad (47)$$

For the second equality, use

$$f(z) = C + \frac{1}{\pi} \int_0^1 \frac{\omega(x)}{x - z} dx$$

and subtract $D_0 f$ from (43), then let $z \rightarrow 0-$. Equation (45) gives $\mathcal{D}(x) - D_0 = O(x^{3/4})$, so the integrand after subtraction is $O(x^{-1/2})$ at zero. This ensures integrability and justifies the limit. On the left, terms involving $f(z)(\tilde{\mathcal{J}}(z) - D_0)$ and $f(z)^2 \tilde{\mathcal{E}}(z)$ tend to zero.

The moment equations now have a unique solution. It remains to exhibit their probabilistic solution, thereby identifying G with the expectation in (40). Let $\mathcal{W}_n$ denote the product in (39) with n coordinates, viewed on $(0, 1)^n$ without the ordering constraint. We use its normalized symmetric law for $n = N$. Partial fractions give

$$\prod_{i=1}^N \frac{y_i - x}{y_i + x} = 1 - x \sum_{i=1}^N \frac{\alpha_i}{x + y_i}, \quad \alpha_i = 2 \prod_{j \neq i} \frac{y_j + y_i}{y_j - y_i}.$$

After multiplication by the density, the factors in α_i cancel one Vandermonde and all corresponding denominator factors. Each canceled term is integrable: its remaining polynomial factors and pair kernels are bounded on $(0, 1)^N$, as is $x/(x + y_i)$, and the one-variable weights are integrable. Integrating over the other variables therefore gives a representation $1 - x \int_0^1 P(y) \omega(y)/(x + y) dy$ with $\deg P \leq N - 1$. Its moments against $x^k \omega(x)$ vanish for $k < N$. To see this, adjoin x as an $(N + 1)$ st variable and use the density identity

$$x^k \omega(x) \mathcal{W}_N(y_1, \dots, y_N) \prod_{i=1}^N \frac{y_i - x}{y_i + x} = \frac{x^k \mathcal{W}_{N+1}(x, y_1, \dots, y_N)}{\prod_{i=1}^N (y_i - x)}.$$

The left side is absolutely integrable, since the product has modulus at most one. Symmetrizing the right side in all $N + 1$ variables gives zero by the elementary interpolation identity

$$\sum_{r=0}^N \frac{z_r^k}{\prod_{j \neq r} (z_j - z_r)} = 0, \quad 0 \leq k < N.$$

Thus the expectation has the same polynomial Cauchy representation and vanishing moments as G . The uniqueness argument using (46) now identifies this expectation with G . Substitute into (47). $\square$

6 The smallest coordinate and the strip exponent

The positive integral reduces the growth of the arch moment to the coordinates nearest zero. We prove the required bounds by comparing with an explicitly normalized Bures–Laguerre density. The comparison is at the level of positive measures, so it also controls the possible sign changes in the product appearing inside the integral.

The integral formula immediately gives

$$\mathbb{E}y_1^{-1/4} \lesssim m_N \lesssim \mathbb{E} \left(\sum_{i=1}^N y_i^{-1} \right)^{1/4}. \quad (48)$$

To verify both estimates pointwise, put $R(x) = \prod_i (y_i - x)/(y_i + x)$. Since each factor lies in $[-1, 1]$, $1 - R(x) \geq 0$, and telescoping the product gives

$$1 - R(x) \leq 2 \min \left\{ 1, x \sum_i y_i^{-1} \right\}.$$

For $y_1/3 < x < y_1/2$, all factors are nonnegative and the first is at most $1/2$, so $1 - R(x) \geq 1/2$. Finally, $\omega(x) \lesssim x^{-1/4}$ on $(0, 1)$ and $\omega(x) \asymp x^{-1/4}$ on $(0, 1/2)$. Integrating the lower bound on the indicated interval and splitting the upper integral at $x = (\sum_i y_i^{-1})^{-1}$ proves (48).

The reference inverse-moment estimates below will contribute a factor $N^{1/2}$. Scaling those coordinates by a factor of order N^{-1} will contribute a further $N^{1/4}$, producing the required $N^{3/4}$ for the arch moment. We now prove both the reference estimates and the comparison that permits this scaling.

For comparison we use the generalized Bures–Laguerre family [12], in ordered coordinates and with parameter $a > 0$, having density

$$\frac{1}{Z_n(a)} \prod_{i=1}^n r_i^{a-1} e^{-r_i} \prod_{i < j} \frac{(r_j - r_i)^2}{r_i + r_j}, \quad 0 < r_1 < \dots < r_n < \infty. \quad (49)$$

Expectations and probabilities for this law will carry subscripts n, a when clarification is needed. The exponent of the one-variable weight is $a - 1$ in our convention; thus our a equals the parameter in [12, equation (1.5)] plus one. Our ordered normalizer is the corresponding unordered integral divided by $n!$. The following association statement is the continuous MTP₂ principle of Sarkar, for log-supermodular densities [4, Theorem 2.27]; see also [10, equation (3.5)]. We give a direct conditioning proof for the ordered pair kernel used here.

Lemma 6.1 (Association and monotone comparison). *Let $I \subset (0, \infty)$ be an interval, let $n \geq 1$, and let $w_1, \dots, w_n$ be positive smooth functions on I . Suppose*

$$\prod_{i=1}^n w_i(x_i) \prod_{i < j} K(x_i, x_j), \quad K(x, y) = \frac{(y - x)^2}{x + y}, \quad x_1 < \dots < x_n, \quad x_i \in I,$$

has finite positive normalizing integral. Its probability law is positively associated: increasing integrable functions have nonnegative covariance whenever the covariance is defined. The assertion also holds after truncation to a shorter common interval. Consequently, a coordinatewise increasing likelihood ratio shifts the coordinates upwards in stochastic order, and a decreasing one shifts them downwards.

Proof. The mixed derivative of each pair contribution is

$$\frac{\partial^2}{\partial x \partial y} \log K(x, y) = \frac{2}{(y-x)^2} + \frac{1}{(x+y)^2} > 0.$$

We prove association by induction on the number of coordinates, initially for bounded increasing functions. Conditional on the largest coordinate t , the remaining coordinates have a law of the same form on the shorter interval, with modified one-variable factors. They are associated by induction. If $s < t$, the ratio of the conditional density at s to that at t , on the former support, is coordinatewise decreasing by the mixed-derivative inequality. Extended by zero where a coordinate exceeds s , it remains decreasing on the latter support. Association for the conditional law at t shows that a decreasing likelihood tilt decreases every increasing expectation. If the ratio is unbounded, first truncate it and pass to the limit by monotone convergence. Thus the conditional expectation of any increasing function of all the coordinates increases with t .

The law of total covariance now has two nonnegative terms: conditional covariance by induction, and covariance of the two conditional expectations by the one-dimensional inequality

$$\text{Cov}(F(T), G(T)) = \frac{1}{2} \mathbb{E}[(F(T) - F(T'))(G(T) - G(T'))] \geq 0.$$

Here T' is an independent copy of T . This proves association; truncation and integrable approximation extend the assertion to the stated class. Finally for an increasing likelihood ratio L , $\mathbb{E}[FL]/\mathbb{E}L \geq \mathbb{E}F$ for increasing F , and the reversed inequality holds for decreasing L . This is the claimed stochastic comparison. $\square$

The upper comparison will use $a = 1/2$, and the lower comparison will use $a = 1$. To estimate the minimum for $a = 1$, we will first need an inverse-moment bound for $a = 2$. The next lemma develops all three estimates from the same normalizing integral.

Lemma 6.2 (Laguerre normalization and coordinate bounds). *For fixed $a > 0$ and every integer $n \geq 1$,*

$$Z_n(a) = \prod_{i=0}^{n-1} \Gamma(a+i) \prod_{0 \leq i < j < n} \frac{j-i}{2a+i+j}. \quad (50)$$

There exist constants K_a, C_a , depending only on a , such that $\mathbb{P}_{n,a}(r_n \leq K_a n)$ is bounded below by a positive constant independent of n , and, with $D_k = ak + k(k-1)/2$,

$$\mathbb{P}_{n,a}(r_k < h) \leq \min \left\{ 1, \left(\frac{C_a h n^2}{k^3} \right)^{D_k} \right\}, \quad h > 0, \quad 1 \leq k \leq n. \quad (51)$$

In particular,

$$\mathbb{E}_{n,1/2} \left(\sum_i r_i^{-1} \right)^{1/4} \lesssim n^{1/2}, \quad (52)$$

$$\mathbb{E}_{n,2} \sum_i r_i^{-1} \lesssim n^2, \quad (53)$$

$$\mathbb{E}_{n,1} r_1^{-1/4} \gtrsim n^{1/2}. \quad (54)$$

The constants in the last three bounds are absolute.

Proof. Normalization. Write the pair product as a Vandermonde determinant times the Schur Pfaffian with entries $(s - r)/(s + r)$. The classical identity, in the form recorded in [19, equation (1.2)], is

$$\text{Pf} \left[\frac{x_j - x_i}{x_j + x_i} \right] = \prod_{i < j} \frac{x_j - x_i}{x_j + x_i}.$$

For even size, clearing denominators leaves an alternating polynomial of Vandermonde degree, so only its constant factor needs checking. Pairing two opposite arguments gives that constant recursively as one. For odd size, add an argument tending to infinity; the resulting augmenting column consists of ones.

Expand the determinant and the Pfaffian and integrate over the unordered region, divided by $n!$. This is de Bruijn's determinant–Pfaffian integration formula [5]; the general antisymmetric-kernel form, including its odd-size border, is stated in [1, Theorem 6.1]. The result is the Pfaffian of double moments, with the single moments as augmenting entries in odd size. Absolute integrability follows from $|(s - r)/(s + r)| \leq 1$ and the finite gamma moments for $a > 0$. The double moment with indices i, j is

$$\int_0^\infty \int_0^\infty r^{a+i-1} s^{a+j-1} e^{-r-s} \frac{s-r}{s+r} dr ds = \Gamma(a+i)\Gamma(a+j) \frac{j-i}{2a+i+j}.$$

Use $r+s$ and $r/(r+s)$ to evaluate the integral; the latter beta integral has mean $(a+i)/(2a+i+j)$. Factoring out the gamma terms and using the Schur identity again, now at the arguments $a+i$, proves (50).

Coordinate tails and inverse moments. For the maximum, condition on its value s . Every factor involving a smaller coordinate is at most s , since $K(r, s) \leq s$ for $0 < r < s$. Consequently

$$\mathbb{P}_{n,a}(r_n > Kn) \leq \frac{Z_{n-1}(a)}{Z_n(a)} \int_{Kn}^\infty s^{a+n-2} e^{-s} ds.$$

The explicit normalizer gives

$$\frac{Z_{n-1}(a)}{Z_n(a)} = \frac{1}{\Gamma(a+n-1)} \prod_{j=1}^{n-1} \frac{2a+2n-2-j}{j} \leq \frac{e^a (2e)^{n-1}}{\Gamma(a+n-1)}.$$

For $n \geq 2$, the inequality follows by replacing every numerator by $2(a+n-1)$, using $(n-1)! \geq ((n-1)/e)^{n-1}$, and then $(1+a/(n-1))^{n-1} \leq e^a$; $n=1$ is immediate. The exponential-moment bound for a gamma random variable gives

$$\frac{1}{\Gamma(a+n-1)} \int_{Kn}^\infty s^{a+n-2} e^{-s} ds \leq e^{-Kn/2} 2^{a+n-1}.$$

Choosing K large makes the desired tail uniformly small, and in fact exponentially small in n .

For the small-coordinate estimate, integrate the first k coordinates below h . Their mutual pair factors are at most h ; a factor between a small coordinate and a remaining one is at most the latter. Discarding order constraints between these two groups therefore gives

$$\mathbb{P}_{n,a}(r_k < h) \leq \frac{h^{D_k}}{a^k k!} \frac{Z_{n-k}(a+k)}{Z_n(a)}, \quad (55)$$

with $Z_0 = 1$. From (50),

$$\frac{Z_{n-k}(a+k)}{Z_n(a)} = \prod_{i=0}^{k-1} \Gamma(a+i)^{-1} \prod_{\substack{0 \leq i < j < n \\ i < k}} \left(1 + \frac{2a+2i}{j-i} \right).$$

For gaps $j - i \leq k$, use $2a + 2i \leq C_a k$. For each $i < k$,

$$\sum_{j=1}^k \log(1 + C_a k/j) \leq k \log(1 + C_a) + k \log k - \log(k!) = O_a(k).$$

Thus the short gaps contribute $O_a(k^2) = O_a(D_k)$. For larger gaps, $\log(1+x) \leq x$ gives a contribution at most

$$\sum_{i < k} (2a + 2i) \sum_{j=k+1}^{n-1-i} \frac{1}{j} \leq 2D_k \log(n/k).$$

This includes $k = n$, when all the latter sums are empty. Furthermore

$$\prod_{i=0}^{k-1} \Gamma(a + i) \geq k^{D_k} e^{-O_a(D_k)}. \quad (56)$$

Here is a direct estimate. For $q = a + i$, integration on $[q, q + 1]$ gives $\log \Gamma(q) \geq (q - 1) \log q - q - O_a(1)$. Summing, separating the factor $\log k$, and using $\sum_{i < k} (a + i) \log((a + i)/k) \geq -k^2/e$ gives $D_k \log k - O_a(D_k)$; the missing $k \log k$ is absorbed in $O_a(D_k)$. Adjust the constant for bounded k . Combining these bounds with (55), and absorbing $a^{-k}/k!$ into $C_a^{D_k}$, proves (51).

For any $0 < p < D_k$, integration of that tail estimate gives

$$\mathbb{E}_{n,a} r_k^{-p} \leq \frac{D_k}{D_k - p} \left(\frac{C_a n^2}{k^3} \right)^p. \quad (57)$$

At $a = 1/2$, apply this with $(k, p) = (1, 1/4)$ and then with $p = 1, k \geq 2$. It gives $\mathbb{E} r_1^{-1/4} \lesssim n^{1/2}$ and $\mathbb{E} \sum_{k \geq 2} r_k^{-1} \lesssim n^2$. Subadditivity of the fourth root and Jensen's inequality yield (52). At $a = 2$, use $p = 1$ for every k to obtain (53).

The minimum at $a = 1$. The density of the smallest coordinate at $s > 0$ is

$$e^{-s} \frac{Z_{n-1}(2)}{Z_n(1)} \mathbb{E}_{n-1,2} \left[\mathbf{1}_{\{r_i > s \ \forall i\}} \prod_i \frac{(1 - s/r_i)^2}{1 + s/r_i} \right].$$

The normalizer ratio is exactly $n(n+1)/2$. Set $q_s(r) = \mathbf{1}_{\{r > s\}}(1 - s/r)^2/(1 + s/r)$. For all $r > 0$, including $r \leq s$, one has $0 \leq q_s(r) \leq 1$ and $1 - q_s(r) \leq 3s/r$. Consequently

$$\prod_i q_s(r_i) \geq 1 - 3s \sum_i r_i^{-1}.$$

By (53), the expectation is at least $1/2$ whenever $0 < s < \varepsilon n^{-2}$ for a sufficiently small fixed $\varepsilon > 0$. The displayed density is then bounded below by a positive multiple of n^2 . An interval with endpoints fixed positive multiples of n^{-2} has probability bounded below; on it $r_1^{-1/4} \gtrsim n^{1/2}$. This proves (54), with $n = 1$ also covered directly. $\square$

Completion of Theorem 1.1. We compare (39) with the reference laws to estimate the two sides of (48). For the upper estimate, first condition the target law on $y_N < 1/2$. This decreasing event shifts all coordinates downwards, by Lemma 6.1, and therefore increases the inverse statistic. On this shorter interval compare it with $y_i = r_i/(bN)$ under (49) with $a = 1/2$, conditioned on $r_N < bN/2$. Up to a constant, the target-to-reference likelihood ratio is the product of

$$\omega(y) y^{1/2} e^{bNy}.$$

Direct differentiation gives

$$\frac{d}{dy} \log(\omega(y)y^{1/2}) = \frac{1 - 2y - \sqrt{2y/(1+y)}}{4y(1-y)} \geq -4, \quad 0 < y \leq \frac{1}{2}.$$

Indeed the numerator is positive for $y \leq 1/8$; on the remaining interval it is at least -1 and the denominator is at least $1/4$. Thus the likelihood factors are increasing for every $N \geq 1$ once $b \geq 4$. Association shows that the conditioned reference law has the larger inverse statistic. Choosing also $b/2 \geq K_{1/2}$ makes its conditioning probability bounded below uniformly in N , by Lemma 6.2. Hence

$$\mathbb{E} \left(\sum_i y_i^{-1} \right)^{1/4} \lesssim (bN)^{1/4} \mathbb{E}_{N,1/2} \left(\sum_i r_i^{-1} \right)^{1/4} \lesssim N^{3/4}.$$

For the lower estimate, set $y_i = 2z_i/(1+z_i)$ in the target law, so $0 < z_1 < \dots < z_N < 1$. The transformed pair kernel is

$$K \left(\frac{2z_i}{1+z_i}, \frac{2z_j}{1+z_j} \right) = \frac{2K(z_i, z_j)}{(1+z_i)(1+z_j)(1+2z_i z_j/(z_i+z_j))}.$$

Together with the Jacobian this gives one-variable factors $\omega(2z/(1+z))(1+z)^{-(N+1)}$ and additional pair factors $(1+2z_i z_j/(z_i+z_j))^{-1}$. Compare with $z_i = r_i/(bN)$ under the reference law with $a = 1$ and fixed $0 < b < 1/2$, conditioned on $r_N < bN$. The ratio has individual factors

$$\omega \left(\frac{2z}{1+z} \right) (1+z)^{-(N+1)} e^{bNz}$$

and the additional pair factors just displayed. Each is decreasing in every coordinate: ω is decreasing, and $-(N+1)/(1+z) + bN < 0$ for $0 < z < 1$. Association therefore puts the target z coordinates below those of this conditioned reference, which are themselves below the unconditioned reference coordinates. Since $y_1 \leq 2z_1$, it follows that

$$\mathbb{E} y_1^{-1/4} \geq 2^{-1/4} (bN)^{1/4} \mathbb{E}_{N,1} r_1^{-1/4} \gtrsim N^{3/4}.$$

Equation (48) now proves $m_N \asymp N^{3/4}$.

It remains to extract the bridge estimate from Lemma 4.2. By monotonicity, summing the lower increment bound over $1 \leq j < N$ gives $(N-1)\mathcal{B}_N \lesssim m_N$, so $\mathcal{B}_N \lesssim N^{-1/4}$, with bounded N absorbed in the constant. Conversely choose a fixed integer L so large that the lower bound for m_{LN} exceeds twice the upper bound for m_N . Then

$$N^{3/4} \lesssim m_{LN} - m_N \lesssim \sum_{j=N}^{LN-1} \mathcal{B}_j \leq (L-1)N\mathcal{B}_N.$$

This proves $\mathcal{B}_N \gtrsim N^{-1/4}$ and completes all assertions of the theorem. $\square$

A Local coefficient identities

This appendix proves Lemma 2.2 in the diagram convention of Section 2. The finite tables also specify every nonzero local coefficient used by the interpolation argument.

Proof. Equation (8) follows by direct substitution. For completeness we record a coefficient check of the splitting identity. Subscripts 1, 2 mean arguments $x + s\lambda, x$. Let q be the cup coefficient and k the strand coefficient of the splitter: $q = k = \rho$ for $s = 2$, and $q = 1, k = 0$ for $s = 3$. Unsubscripted weights refer to $\hat{R}_s$; in the deletion case use $B = 1$ and set its other nonempty two-slot coefficients to zero. The inputs are the passing auxiliary strand and the slot to be split; the latter is absent in the deletion case. Outputs 1, 2 are the split slots and output 3 is the auxiliary slot. Input bits record occupancy. For one input strand, a bare output index specifies its destination and a parenthesized pair specifies an additional cup. The complete coefficient check is

input	connections at output	left side	$(S_s \otimes I)\hat{R}$
00	$\emptyset$	1	1
	12 (cup)	$U_1 A_2 + q B_1 B_2$	q
	13	$U_1 B_2 + q B_1 A_2$	kU
	23	$U_2 + q A_1 E_2$	kU
10	1	$A_1 + q F_1 U_2$	kA
	2	$B_1 A_2 + q U_1 B_2$	kA
	3	$B_1 B_2 + q U_1 A_2$	B
	3, (12)	$q F_1 E_2$	qB
	1, (23)	$A_1 U_2 + q(E_1 E_2 + F_1 F_2)$	0

For $s = 2$ there are also the following inputs. In the final 11 block, an empty output or a parenthesized output pair connects the two input strands to one another; a bare pair of output indices denotes two through-strands.

01	1	$k(B_1 + U_1 U_2)$	kB
	2	$k(A_1 A_2 + B_2)$	kB
	3	$k(A_1 B_2 + A_2)$	A
	1, (23)	$k(B_1 U_2 + U_1 F_2)$	0
	3, (12)	$k U_1 E_2$	qA
11	$\emptyset$	$k(U_1 + B_1 U_2)$	U
	12	$k(E_1 A_2 + A_1 B_2)$	0
	(12)	$k F_1 A_2$	qU
	13	$k(E_1 B_2 + A_1 A_2)$	kE
	(13)	$k F_1 B_2$	kF
	23	$k B_1 E_2$	kE
	(23)	$k(U_1 U_2 + B_1 F_2)$	kF

Here are the substitutions verifying these equalities. Put $s_j = \sin(x + j\lambda)$. The weights at $x + j\lambda$ have common denominator $s_{j+2}s_{j+3}$ and numerators

weight	A	B	U	E	F
numerator	ds_{j+5}	$s_j s_{j+5}$	ds_j	$s_{j+5} s_{j+6}$	$-s_{j+7} s_j$

Substitution uses the following product-to-sum identities:

$$\begin{aligned}
s_{j+8} &= -s_j, & s_j s_{j+1} + s_{j+4} s_{j+5} &= C, & s_2 s_6 &= d^2 - s_0^2, & s_1 s_5 &= d^2 - s_7^2, \\
\rho(s_1 - s_7) &= s_0, & \rho(s_6 - s_0) &= s_7, & \rho(s_1 s_2 - s_6 s_7) &= s_0 s_4, & \rho(s_5 s_6 - s_0 s_1) &= s_7 s_3, \\
\rho(s_7 s_0 + s_4 s_5) &= s_6 s_2, & \rho(s_2 s_3 + s_7 s_0) &= s_1 s_5, & s_1 s_6 - s_7 s_0 &= \rho d^2, \\
s_1 s_6 s_5 &= s_7 s_2 s_3 + d^2 s_0, & s_1 s_6 s_2 &= s_0 s_4 s_5 + d^2 s_7
\end{aligned}$$

The last two trigonometric identities also follow from the preceding formulas for s_1s_5 and s_6s_2 . Thus substitution verifies every entry in the tables.

For (10), the sector with one strand reduces to multiplication of two matrices with diagonal entries A and off-diagonal entries B . In the even sector the required identities are

$$\begin{aligned} E(x)E(-x) &= 1, & U(x) + E(x)U(-x) &= 0, \\ E(x)F(-x) + F(x)E(-x) + U(x)U(-x) &= 0. \end{aligned}$$

A cap composed with a cup creates a loop and therefore vanishes. These identities again follow from (6).

To prove (7), first set $y = s\lambda$, $s = 2, 3$. Use (8) and (9), together with their transposes and the reversal of slot order. The effective box is symmetric, and both sides become $(I \otimes S_s)\widehat{R}_s(x)(S_s^t \otimes I)$. Unitarity transports these identities to $x + y = s\lambda$, by applying the identity with arguments $(-x, x + y)$. The case $y = 0$ follows from $R(0) = I$.

Multiply the difference of the two sides by their three scalar denominators, and call the resulting diagram sum $\mathcal{F}(x, y)$. Each coefficient is a Laurent polynomial in e^{iy} with exponents between -4 and 4 . To see its parity, let D_j multiply a diagram by -1 when slot j is occupied. Inspection of the nine boxes gives

$$R_i(v + \pi) = D_{i+1}R_i(v)D_i.$$

On either side of the braid relation the two internal D_2 signs cancel. The remaining signs are at input slot 1 and output slot 3; the scalar denominators are π -periodic. Hence

$$\mathcal{F}(x, y + \pi) = D_3\mathcal{F}(x, y)D_1.$$

Every boundary coefficient has fixed parity. After multiplication by a Laurent monomial it is a polynomial of degree at most four in e^{2iy} . We have proved its vanishing at $y = 0, 2\lambda, 3\lambda, 2\lambda - x, 3\lambda - x$; these five values are distinct modulo π for generic x . Interpolation proves the identity, and rational continuation removes the genericity restriction. $\square$

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